2.2.1 Let $\tau \in \mathcal{H}$ be fixed only by Γ 的单位元 $\in \Gamma$.

use prp 2.1.1 : $\left(\begin{array}{l} \forall \tau_1, \tau_2 \in \mathcal{H} \quad \exists\, U_1 \ni \tau_1,\ U_2 \ni \tau_2 \\ \forall \gamma \in SL_2(\mathbb{Z}) \quad \gamma(U_1) \cap U_2 \neq \emptyset \Rightarrow \gamma(\tau_1) = \tau_2 \end{array} \right)$

to show that some neighborhood $U \ni \tau$ is homeomorphic to its image in $Y(\Gamma)$.

Pf: We need to show $\pi : \overset{\exists U \ni \tau}{U} \xrightarrow{\sim} \Gamma U$ homeomorphic.

By definition of quotient map, π is continuous

Since π is open, π^{-1} is continuous

By definition of quotient map, π is surjective

It remains to show π injective, that is,

$$\Gamma\tau_1 = \Gamma\tau_2 \Rightarrow \tau_1 = \tau_2. \quad \forall \tau_1, \tau_2 \in U$$

Since $\Gamma\tau_1 = \Gamma\tau_2$ we have some $\gamma \in \Gamma$ such that

$$\gamma\tau_1 = \tau_2.$$

Thus $\gamma(U) \cap U \neq \emptyset$. By Prp 2.1.1, $\gamma\tau = \tau$

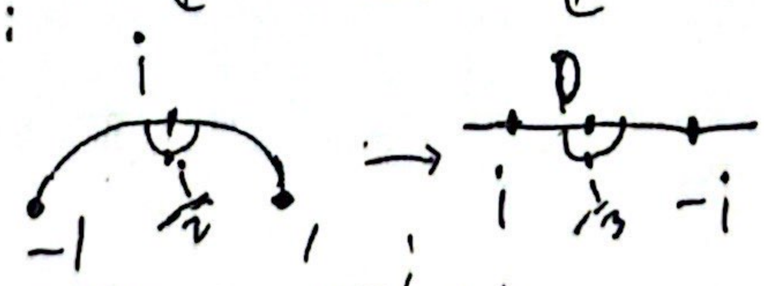
which gives $\gamma = I$ or $-I$ Thus $\tau_1 = \tau_2$.

2.2.2 Compute $\begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix}^{-1}$ and explain $\gamma(\tau) = -\frac{1}{\tau}$ acts in the small as 180° rotation about i

Sol: $\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix} = I \Rightarrow \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{i}{2} & -\frac{i}{2} \end{pmatrix}$.

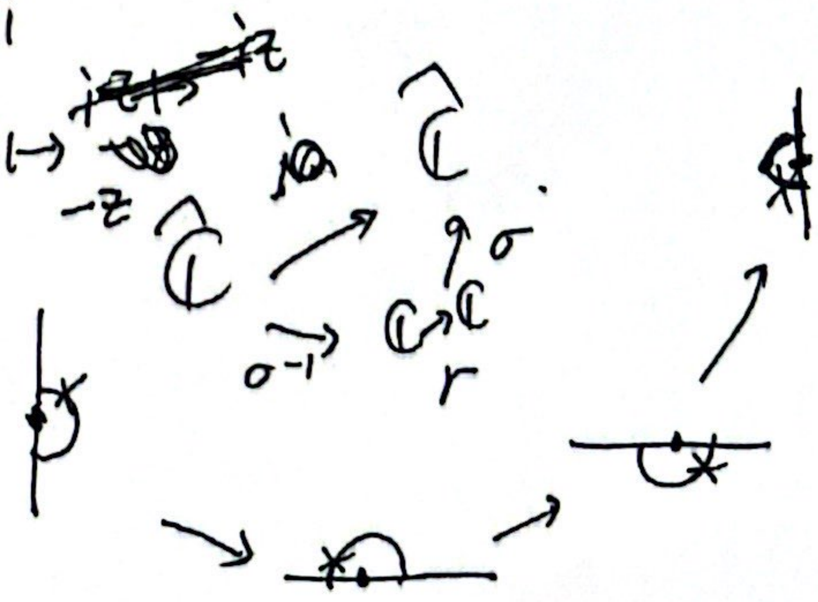
$$\begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix}^{-1} = \begin{pmatrix} -i & -1 \\ i & -1 \end{pmatrix}\begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{i}{2} & -\frac{i}{2} \end{pmatrix} = \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix}$$

Denote $\sigma = \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix} : \mathbb{C} \longrightarrow \hat{\mathbb{C}}$ $\quad z \mapsto \frac{z-i}{z+i}$ $\quad r = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.



Then $\sigma r \sigma^{-1} = \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix} : z \mapsto -z$ in $\hat{\mathbb{C}}$.

Thus $\gamma : \tau \mapsto -\frac{1}{\tau}$ is in $\mathbb{C}$



2.2.3 Show if $\tau \in \mathcal{H}$, $\gamma \in SL_2(\mathbb{Z})$, then $\gamma(\tau)$ under $\gamma\Gamma\gamma^{-1}$ is the same as the period of τ under Γ.

Pf: We need to show $[\{\pm I\}(\gamma\Gamma\gamma^{-1})_{\gamma(\tau)} : \{\pm I\}] = [\{\pm I\}\Gamma_\tau : \{\pm I\}]$

$$\cancel{\{\pm I\}}(\gamma\Gamma\gamma^{-1})_{\gamma(\tau)} = \{\delta \in \gamma\Gamma\gamma^{-1} : \delta(\gamma(\tau)) = \gamma(\tau)\}$$

$$= \{\delta \in \Gamma : (\gamma\delta\gamma^{-1})(\gamma(\tau)) = \gamma(\tau)\} = \{\delta \in \Gamma : \gamma\delta(\tau) = \gamma(\tau)\}$$

$$= \{\delta \in \Gamma : \delta(\tau) = \tau\} = \Gamma_\tau .$$

2.2.4 Prove Cor 2.2.3: Γ congruence subgroup of $SL_2(\mathbb{Z})$.

Each point $\tau \in \mathcal{H}$ has neighborhood $\tau \in U \subset \mathcal{H}$.

s.t $\forall \gamma \in \Gamma$ $\gamma(U) \cap U \neq \phi \Rightarrow \gamma \in \Gamma_\tau$.

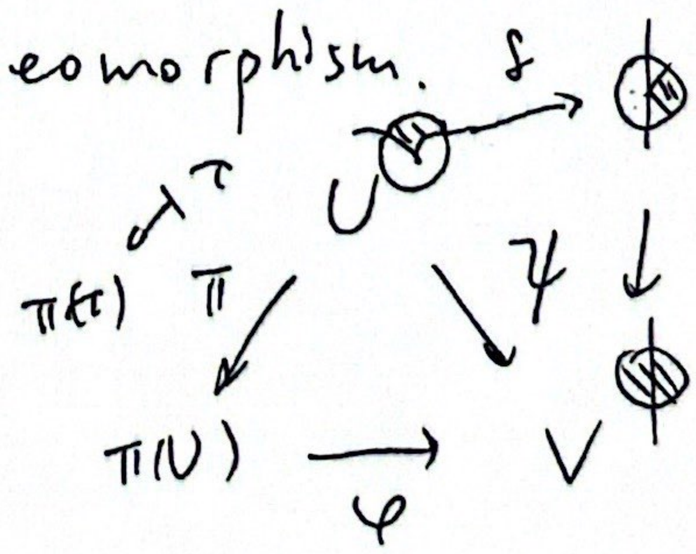
Pf: By Prop 2.1.1, $\forall \tau_1, \tau_2 \in \mathcal{H}$ $\exists \tau_1 \in U_1 \subset \mathcal{H}$
$\tau_2 \in U_2 \subset \mathcal{H}$.

$\forall \gamma \in \Gamma$ $\gamma(U_1) \cap U_2 \neq \phi \Rightarrow \gamma(\tau_1) = \tau_2$

Now let $\tau = \tau_1 = \tau_2$, ~~pick then~~ pick $U_3 = U_1 \cap U_2$.
then $\tau_1 \in U_3 \subset \mathcal{H}$, $\tau_2 \in U_3 \subset \mathcal{H}$.

this gives $\gamma(\tau) = \tau$ i.e. $\gamma \in \Gamma_\tau$.

2.2.5 Explain why $\varphi : \pi(U) \to V$ is homeomorphism.



Pf: $\psi = \varphi \circ \pi$ π open, Continuous.

ψ open continuous.

φ injective, surjective.